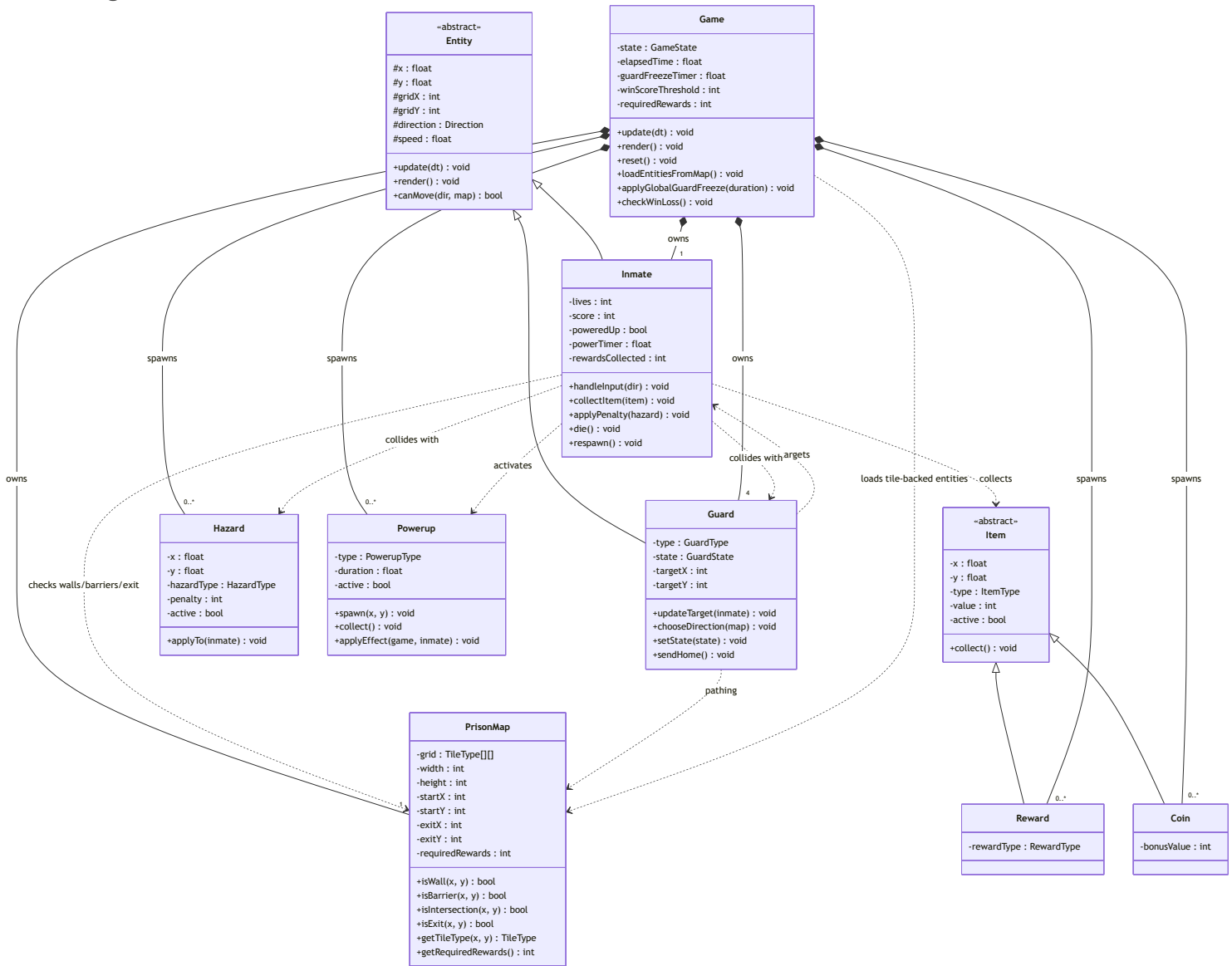
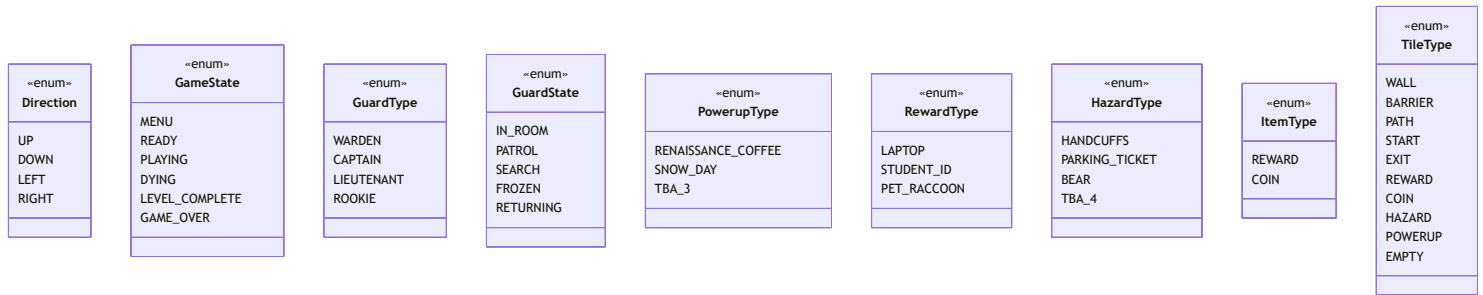


Escape From The Burnaby Mountain Prison (SFU) — UML

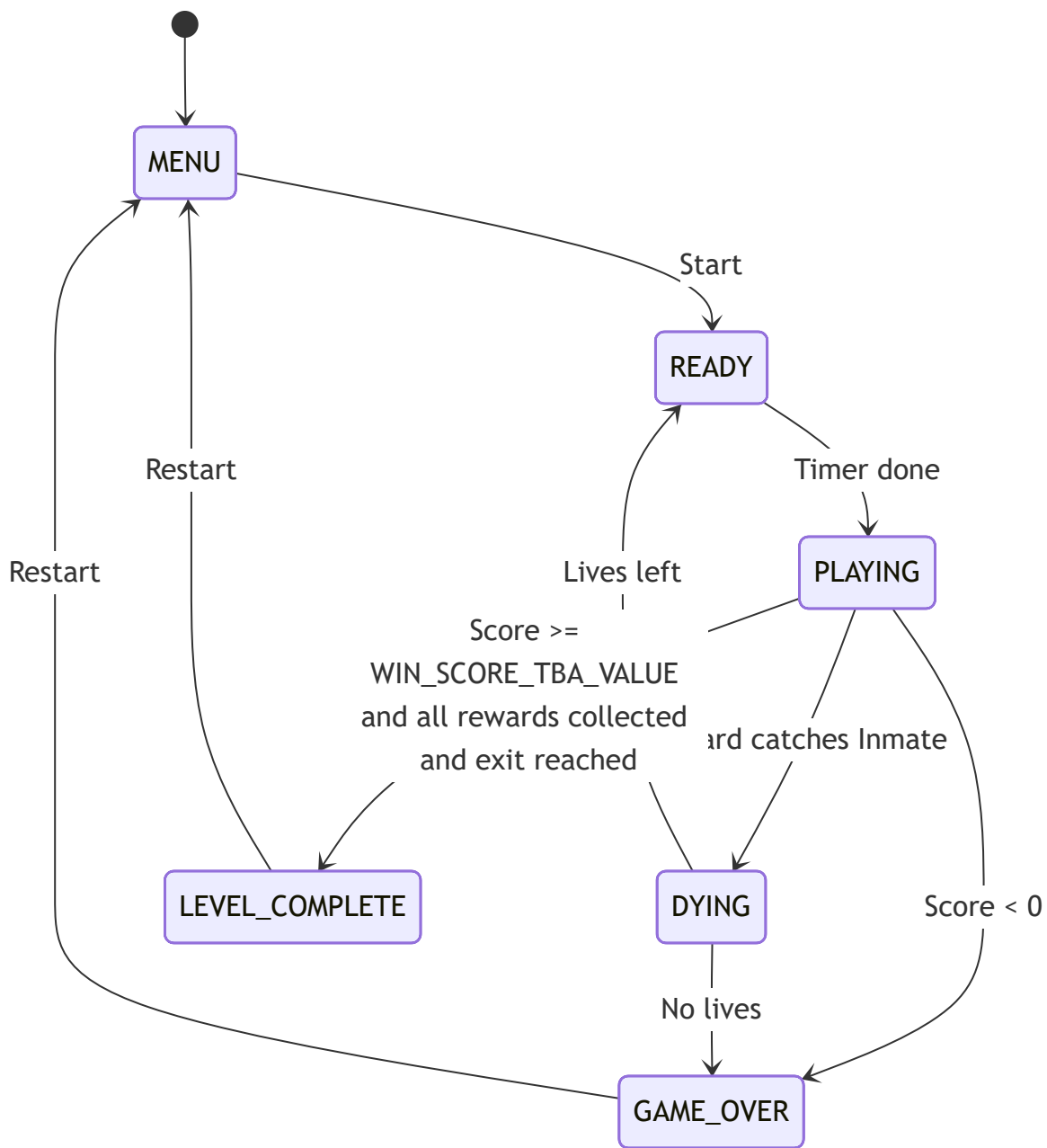
Class Diagram



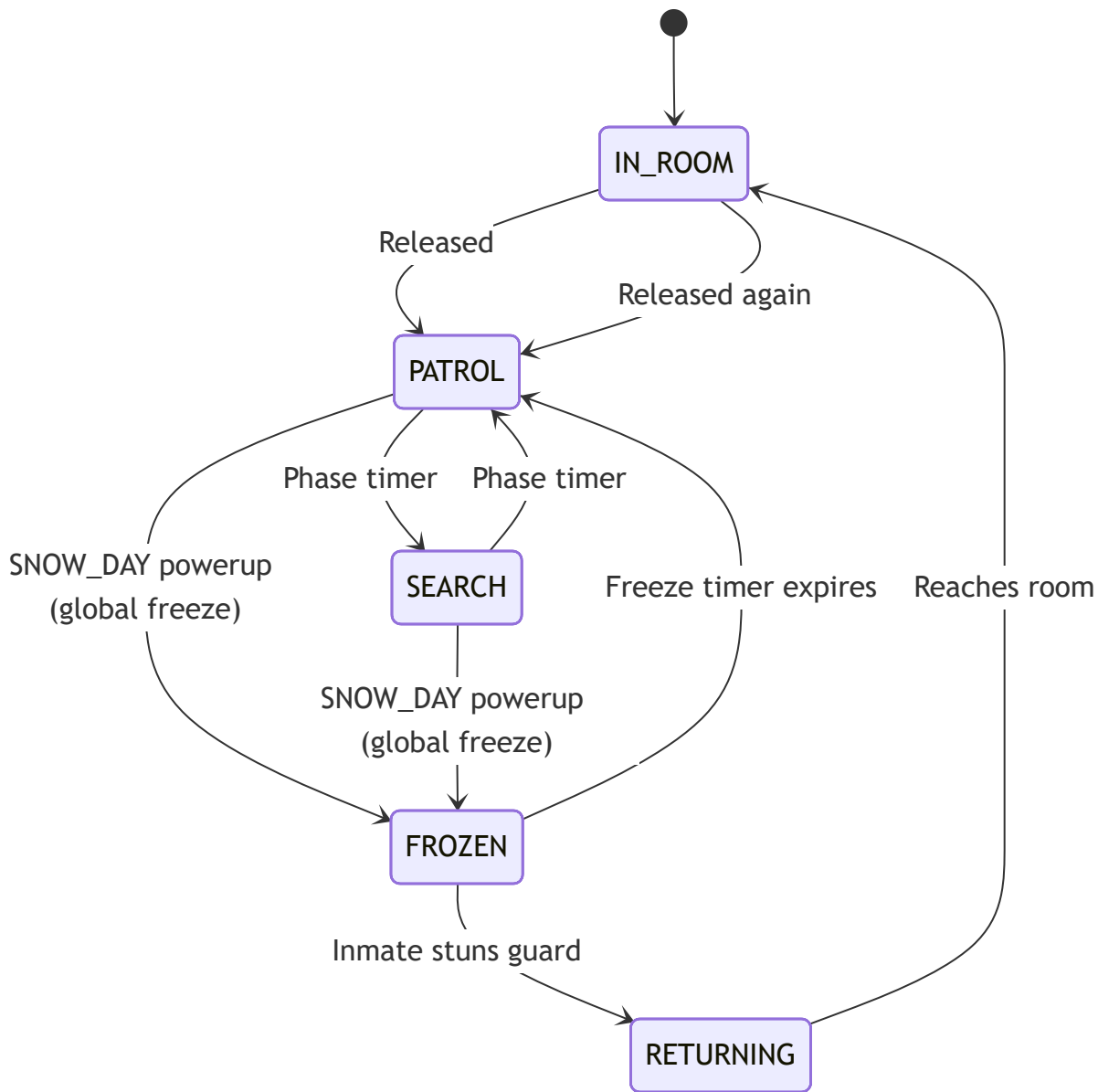
Enums



Game States



Guard States



Key Interactions

- Inmate moves -> checks PrisonMap for walls/barriers
- Inmate on reward tile -> score += reward value, rewardsCollected += 1
- Inmate on coin tile -> score += bonus value
- Inmate on hazard tile -> score -= penalty
- Inmate on RENAISSANCE_COFFEE -> temporary speed increase
- Inmate on SNOW_DAY powerup -> all guards enter FROZEN briefly
- Inmate touches FROZEN guard -> guard RETURNING, score += combo bonus
- Inmate touches normal guard -> Inmate.die(), lives -= 1
- Score < 0 -> game over
- Score >= WIN_SCORE_TBA_VALUE + all required rewards collected + inmate reaches EXIT -> level complete

Guard AI

Guard	SEARCH Target	PATROL Target
Warden	Inmate's tile directly	Top-right corner

Captain	4 tiles ahead of Inmate	Top-left corner
Lieutenant	Flanks using Warden's position	Bottom-right corner
Rookie	Chases if far, retreats if close	Bottom-left corner

PATROL mode: each guard loops to their assigned corner
SEARCH mode: each guard uses their targeting logic above
FROZEN: no movement until global guardFreezeTimer ends, can be stunned
RETURNING: beeline back to guard room

Design Notes

Rewards, coins, hazards, and powerups are tile-backed entities.
Game.loadEntitiesFromMap() materializes object lists from PrisonMap tile markers at startup.
WIN_SCORE_TBA_VALUE is intentionally unresolved and must be set during balancing.